HemoScan AI - Clinical Hematology Report

Hematology Laboratory Report

Patient Identification

* Name: asdcgfvbnm

* Age: 32.0 Years

* Gender: Female

* Phone: 9468213961

* Report Date: October 26, 2023

Complete Blood Count (CBC) Comparison Table

Parameter	Patient Value	Reference Range (Adult Female)	Status
Hemoglobin (Hb)	10.6 g/dL	12.0 – 15.5 g/dL	Low
RBC Count	4.26 x 10^{12}/L	3.80 – 5.20 x 10^{12}/L	Normal
PCV (Hematocrit)	34.2 %	36.0 – 46.0 %	Low
MCV	80.2 fL	80.0 – 100.0 fL	Borderline Low
MCH	25.5 pg	27.0 – 33.0 pg	Low
MCHC	31.8 g/dL	32.0 – 36.0 g/dL	Low

Clinical Interpretation

The laboratory findings confirm a state of anemia, as indicated by the Hemoglobin level of 10.6 g/dL, which falls below the established threshold for adult females. The Packed Cell Volume (PCV) is also reduced at 34.2%, correlating with the decrease in total hemoglobin mass.

A detailed analysis of the red cell indices provides further insight into the morphology of the erythrocytes:

1. MCV (Mean Corpuscular Volume): At 80.2 fL, the average size of the red blood cells is at the absolute lower limit of the normal range. This suggests a transition toward microcytosis (smaller than normal cells).

2. MCH & MCHC: Both the Mean Corpuscular Hemoglobin (25.5 pg) and the Mean Corpuscular Hemoglobin Concentration (31.8 g/dL) are low. This indicates "hypochromia," meaning the red blood cells contain a lower-than-optimal concentration of hemoglobin, often appearing paler under microscopic examination.

The ML model's high confidence score (99.33%) aligns with the clinical data, suggesting a clear pathological deviation from healthy hematological baselines.

Risk Level

Moderate

While a hemoglobin level of 10.6 g/dL is classified as "mild" anemia by WHO standards, the presence of hypochromic indices in a 32-year-old female warrants clinical attention to prevent further decline. If left untreated, this may lead to symptoms such as chronic fatigue, dyspnea (shortness of breath) on exertion, and reduced cognitive focus.

Possible Anemia Type

Microcytic Hypochromic Anemia

The most probable etiology for this presentation is Iron Deficiency Anemia (IDA). In women of reproductive age, this is frequently secondary to menstrual blood loss or nutritional insufficiency. Other differential diagnoses to consider include:

* Early-stage Anemia of Chronic Disease: Often presents with borderline indices.

* Thalassemia Trait: Though the RBC count is usually higher in Thalassemia, it remains a secondary possibility.

* Sideroblastic Anemia: Less common, but involves impaired heme synthesis.

Suggested Diagnostic Tests

To establish a definitive diagnosis and determine the underlying cause, the following investigations are recommended:

1. Iron Studies (Serum Iron Profile): This should include Serum Ferritin (the most sensitive marker for iron stores), Total Iron Binding Capacity (TIBC), and Transferrin Saturation.

2. Peripheral Blood Smear (PBS): A manual microscopic examination to visualize red cell morphology (confirming microcytosis, hypochromia, or anisocytosis).

3. Reticulocyte Count: To evaluate the bone marrow's regenerative response to the anemia.

4. Stool Examination for Occult Blood: To rule out any gastrointestinal blood loss.

5. Mentzer Index Calculation: To help differentiate between Iron Deficiency and Thalassemia Trait (MCV/RBC ratio). In this case, the index is ~18.8, which points more strongly toward Iron Deficiency (>13).